

**The Waiting Game: A Survival Analysis of Fernando Tatis Jr.'s Most Baffling Power
Drought**

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Business Problem

Power hitters are among the highest-value assets in professional baseball. Home run production drives run scoring, influences game strategy, and commands significant salary investment in free agency. When a premier power hitter goes through an extended home run drought, it raises questions from front offices, fantasy analysts, and broadcast commentators alike: Is this a statistical anomaly that will self-correct, or is something structurally wrong?

Standard box score metrics offer no useful signal in this situation. Raw home run totals are silent until a home run actually occurs, and batting average or on-base percentage cannot distinguish between a hitter who is declining and one who is simply unlucky. Fernando Tatis Jr. illustrates this problem precisely: through 42 games and 180 plate appearances of the 2026 season so far, he had zero home runs despite posting an average exit velocity of 91.6 mph and a hard-hit rate among the league elite.

This project addresses three core business questions:

- How long do home run droughts typically last for high-power hitters, measured in plate appearances?
- Which Statcast metrics best predict drought duration, and how much do changes in those metrics extend expected drought length?
- How statistically anomalous is Tatis' 2026 drought, both relative to his career baseline and after adjusting for his current batted ball profile?

The answers have direct applications for player evaluation, fantasy roster management, and broadcast analysis. Any context in which distinguishing between bad luck and a genuine performance shift carries meaningful value.

Background/History

Survival analysis is a branch of statistics developed to model time-to-event data, with a defining feature being its ability to handle right-censored observations, which are cases where the event has not yet occurred by the end of the observation window. In baseball, a home run drought that is still ongoing is a right-censored observation. While survival analysis has been applied to sports contexts including career duration and injury recovery (Baumer et al., 2021), applying it specifically to home run droughts at the plate appearance level is a novel framing made possible by Statcast's rich batted ball data, available league-wide since 2015 (MLB Advanced Media, 2015).

Statcast introduced contact quality metrics such as exit velocity, launch angle, and the barrel classification, that fundamentally changed player evaluation. A barreled ball meets thresholds for both exit velocity and launch angle that historically produce a combined batting average and slugging percentage above 1.500 (Healey, 2017), and its relationship to home run production is among the strongest in the Statcast era.

Fernando Tatis Jr. debuted in 2019 and peaked at 42 home runs in 2021 before losing his 2022 season to suspension and injury (Baseball Reference, 2026). Since returning in 2023, his home run rate has declined each season, reaching a career low in 2025 before the 2026 drought began.

Data Explanation: Sources

All data for this project was sourced programmatically using the pybaseball Python library, which provides clean access to Statcast pitch-level data via the MLB Baseball Savant API and season-level batting statistics via Fangraphs. No proprietary or restricted data sources were used.

Data Explanation: Preparation

The raw Statcast dataset for Fernando Tatis Jr. returned 12,814 rows covering 2017 through May of 2026, with each row representing a single pitch. The following preparations steps were applied:

- Filtered to regular season games only, using `game_type = 'R'` to exclude spring training and postseason appearances.
- Filtered to 2019 onward to ensure consistent Statcast coverage, as pre-2019 rows had limited spring training data only.
- Reduced to one row per plate appearance by retaining only rows where the `events` column is populated. This field records the outcome only on the final pitch of each plate appearance.
- Derived a barrel indicator from the `launch_speed_angle` classification field, where it had a value of 6 = barrel, and set it to null for non-contact plate appearances.
- Derived a pull direction indicator from hit coordinates (`hc_x < 125` for right-handed batters), set to null for non-contact events.

After filtering, the dataset had 3,122 plate appearances across six seasons (2019, 2020, 2021, 2023, 2024, 2025, 2026). Note that 2022 is absent due to his season-long suspension and injury.

Data Explanation: Drought Episode Construction

Home run drought episodes were constructed by grouping consecutive plate appearances between home run events within each season. Episodes were kept within-season to avoid conflating offseason mechanical adjustments and roster changes with in-season variance. Each episode is characterized by its duration (plate appearances), and event indicator (where 1 = ended

in home run, 0 = censored), and mean covariate values across all plate appearances in the episode. This yielded 159 total episodes: 152 uncensored and 7 right-censored.

Data Explanation: Data Dictionary

Key fields used in modeling:

duration	Number of consecutive plate appearances without a homerun in the episode
event_observed	Binary indicator. 1 if the episode ended in a home run, 0 if right-censored.
mean_exit_velo	Mean exit velocity in mph across batted ball events in the episode.
mean_launch_angle	Mean vertical launch angle in degrees across batted ball events.
barrel_rate	Proportion of batted ball events classified as barreled (<code>launch_speed_angle == 6</code>).
pull_rate	Proportion of batted ball events hit to the pull side (<code>hc_x < 125</code> for RHH).
mean_plate_x	Mean horizontal pitch location at the plate across all plate appearances.
mean_plate_z	Mean vertical pitch location at the plate across all plate appearances.

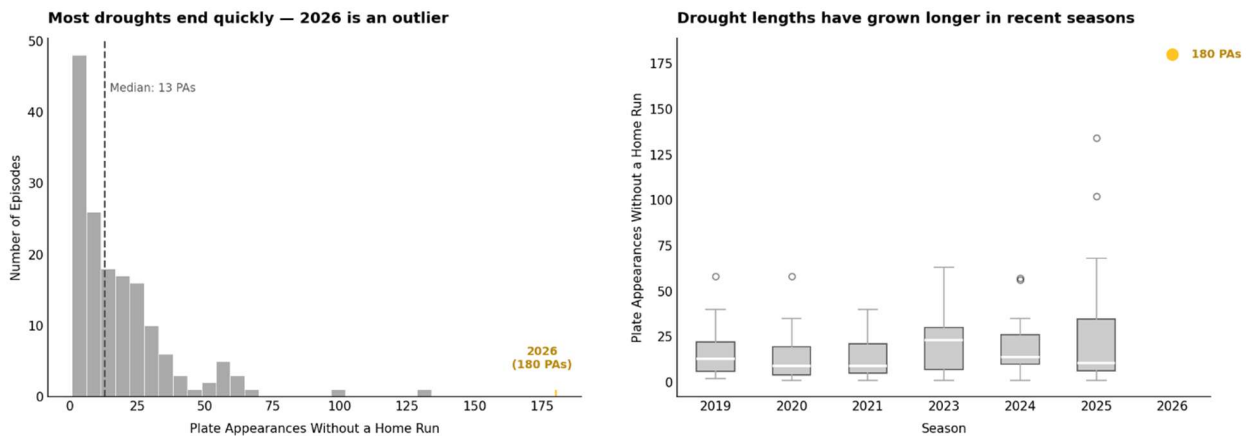


Figure 1: HR Drought Duration Distribution and Season Box Plots

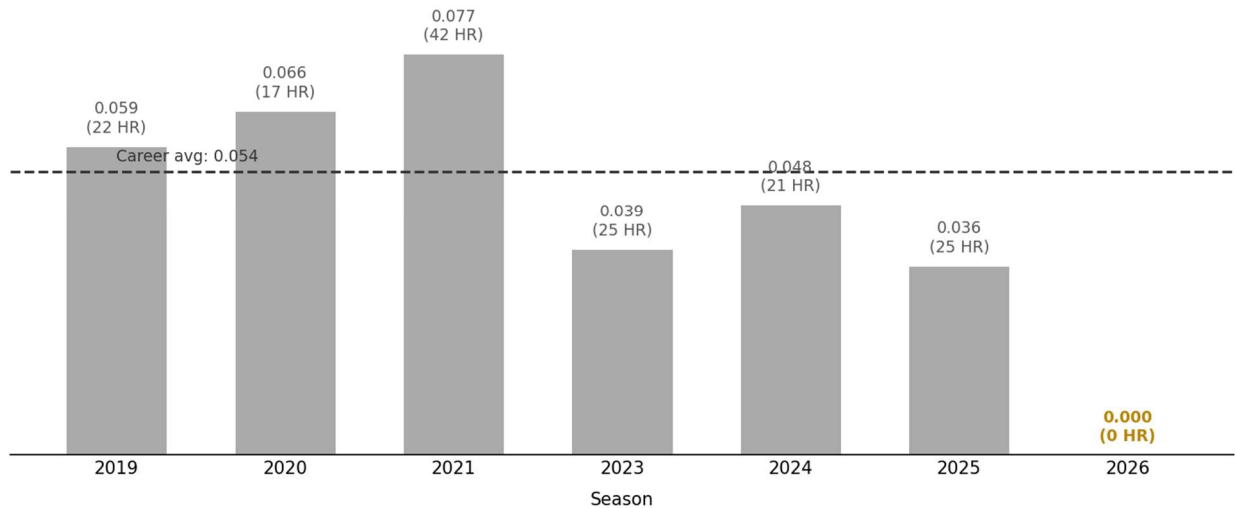
2026 HR rate is zero — a sharp break from career norms

Figure 2: HR Rate per Plate Appearance by Season

Methods

This project applies survival analysis by using two complementary approaches.

Kaplan-Meier Estimator

The Kaplan-Meier (KM) estimator is a non-parametric method that calculates the probability of remaining homer-less at each plate appearance without assuming any underlying time distribution (Kaplan & Meier, 1958). It correctly handles right-censored observations and serves as the primary exploratory tool, establishing baseline survival probabilities before introducing covariates.

Cox Proportional Hazards Model

The Cox Proportional Hazards model is a semi-parametric regression model that estimates the effect of covariates on the hazard, which is the instantaneous probability of hitting a home run at

a given plate appearance, conditional on not having hit one yet (Cox, 1972). It does not require assumptions about the shape of the baseline hazard distribution, making it suited for this application. A ridge regularization penalty (penalizer = 0.1) was applied to stabilize coefficient estimates given the moderate sample size of 158 episodes. The proportional hazards assumption was tested using Schoenfeld residuals.

The final covariate set was selected based on availability across the full date range, theoretical relevance to home run probability, and a pairwise correlation analysis that identified expected wOBA (xwOBA) as redundant with barrel rate and exit velocity ($r = 0.76$) and excluded it from modeling.

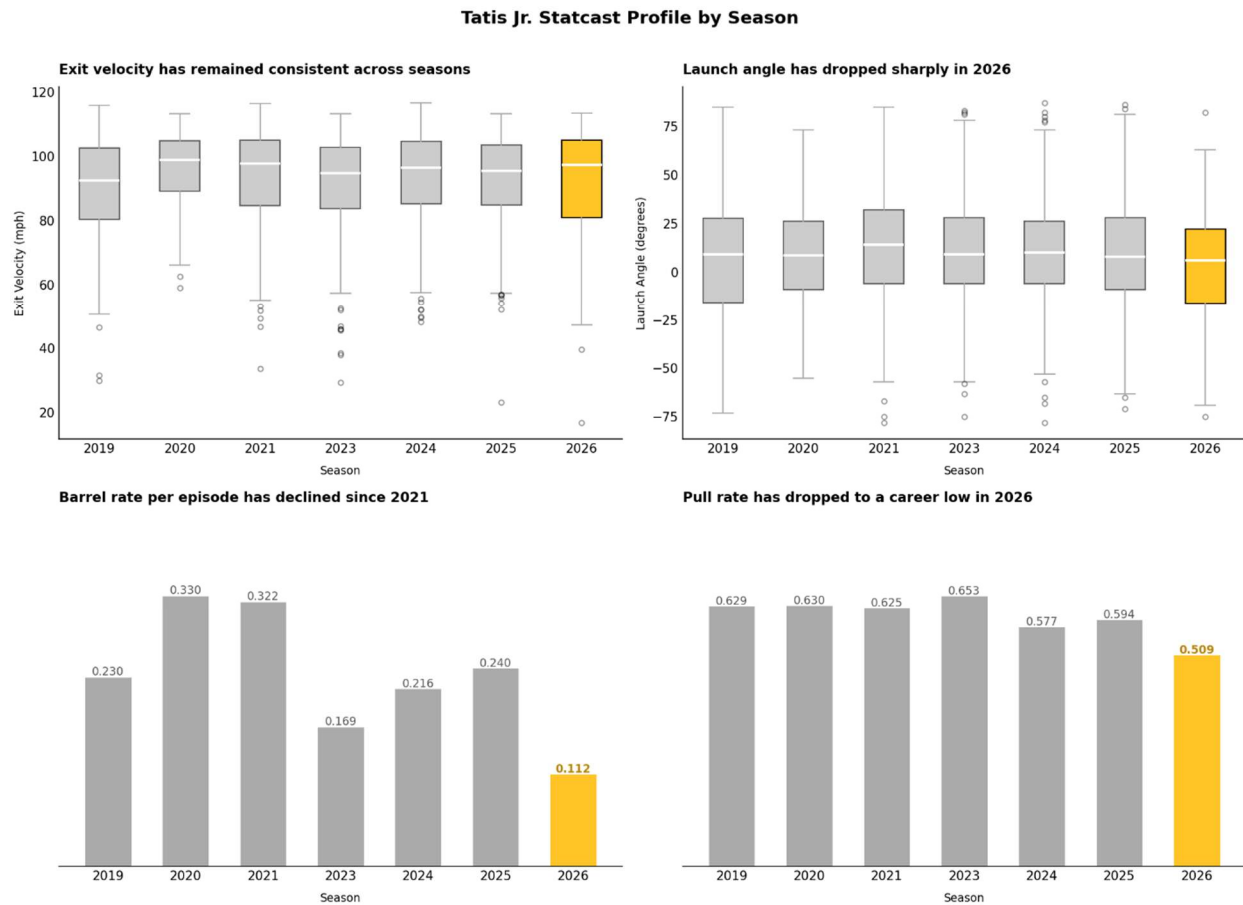


Figure 3: Statcast Covariate Distributions by Season

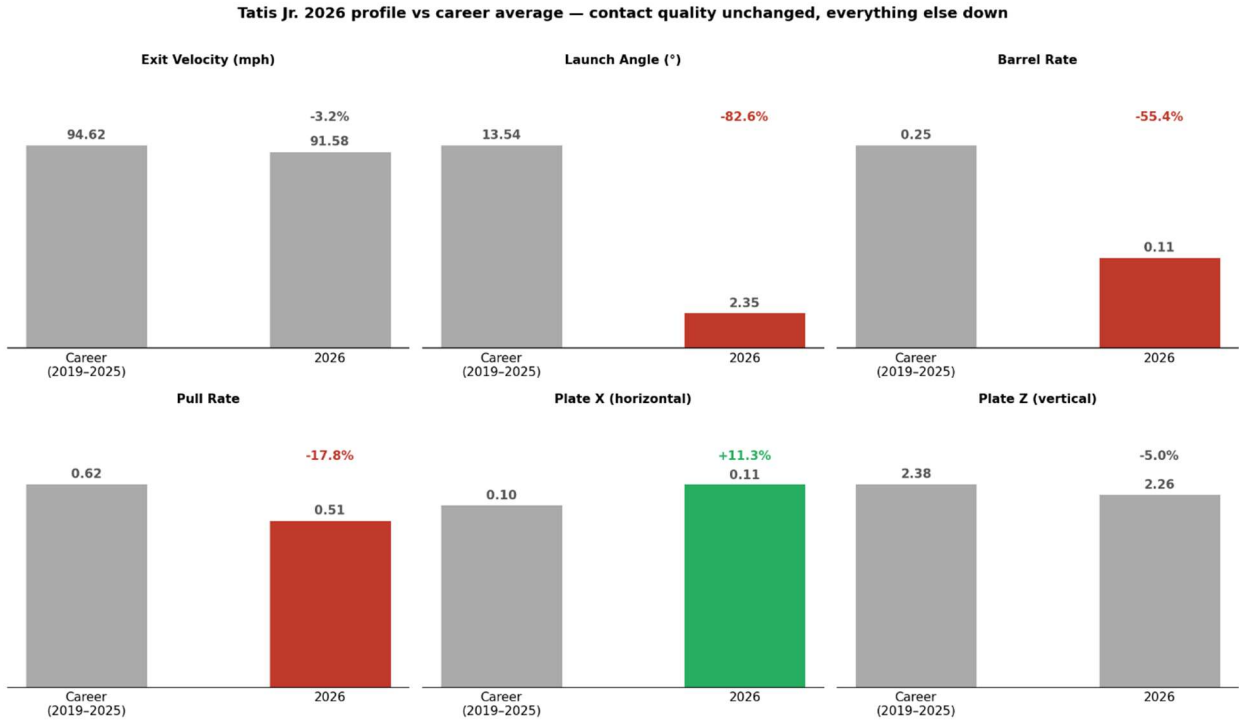


Figure 4: 2026 Profile vs Career Average Comparison

Analysis

The distribution of drought durations is strongly right-skewed, consistent with the expected shape of survival time data. The median drought length across 159 episodes is 13 plate appearances, meaning half of all historical droughts resolved within approximately two weeks of game action. The 2026 drought of 180 plate appearances sits 7.0 standard deviations above the career mean and is visually isolated at the extreme right tail of the distribution.

Examination of covariate trends by season reveals a unified explanation for the drought. Exit velocity has remained stable across Tatis’ career (2026 mean is 91.6 mph, career mean is 94.6 mph, -3.2%). Launch angle has collapsed from a career average of 13.54 degrees to just 2.35 degrees in 2026 (-82.6%). Barrel rate has dropped form a career average of 0.251 to 0.112 (-

55.4%), and pull rate has fallen to a career low of 0.509 (-17.8%). The contrast between stable exit velocity and collapsed launch angle is the defining feature of his 2026 profile.

Kaplan-Meier Results

The KM survival curve confirms that 180 plate appearances (PAs) represents historically rare territory. By 25 PAs, 72.3% of historical droughts have ended. By 50 PAs, 90.5% have ended. The estimated probability of reaching 180 PAs home-less is 0.85%, placing the current drought in the extreme tail of his personal distribution.

Cox Model Results

The Cox model achieved a concordance index of 0.773, correctly ranking episode durations 77.3% of the time. The overall model is highly significant (log-likelihood ratio test, $p < 0.0005$).

Two covariates reached statistical significance:

- Barrel rate (coef=3.854, HR=47.17, $p < 0.00005$): This is a dominant predictor. A one-unit increase in barrel rate increases the hazard of hitting a home run by 47x, consistent with the EDA finding that barreled balls convert to home runs at 50.0% versus 0.5% for non-barreled contact.
- Mean launch angle (coef=0.031, HR=1.031, $p = 0.003$): Each additional degree of average launch angle increases the hazard by approximately 3.1%.

Exit velocity, pull rate, and plate discipline metrics were not individually significant, but all pointed in theoretically consistent directions. The proportional hazards assumption held for all six covariates (all Schoenfeld residual p -values > 0.05).

Predicted Survival Curves and Anomaly Quantification

Applying the fitted Cox model to Tatis' career profile predicts a median drought of 12 plate appearances and a 0.01% probability of reaching 180 PAs. Under his 2026 profile, the predicted median extends to 26 plate appearances and the probability at 180 PAs rises to 3.96%. His changed Statcast profile, defined primarily by the launch angle and barrel rate collapse, more than doubles the expected drought duration and raises the probability of this specific outcome by a factor of roughly 400 compared to his career baseline.

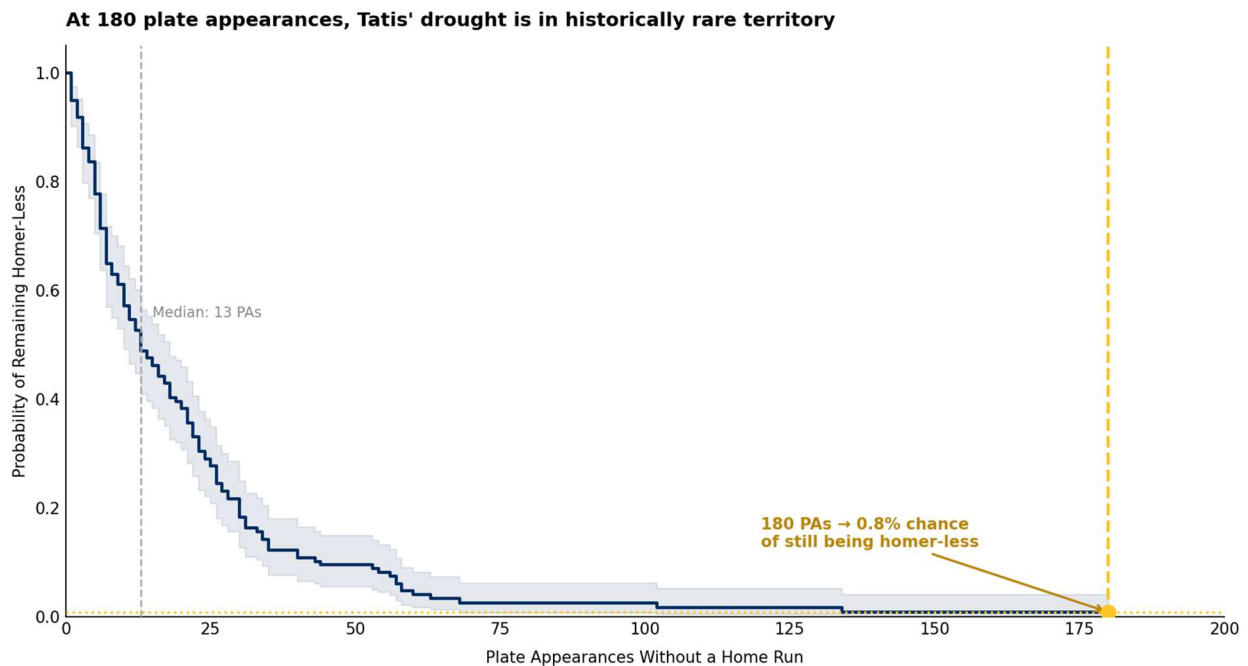


Figure 5: Kaplan-Meier Survival Curve with 2026 Drought Marked

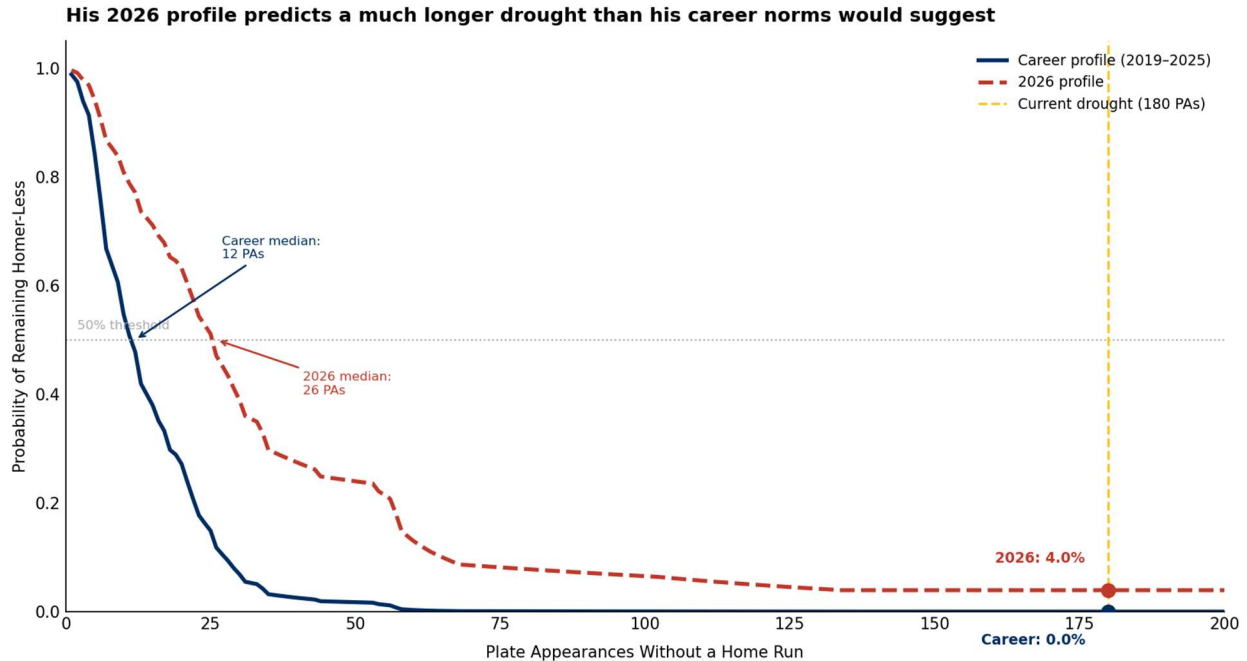


Figure 6: Predicted Survival Curves. Career Profile vs 2026 Profile

Conclusion

Tatis is hitting the ball just as hard as ever. His exit velocity is only down 3.2% from his career average, confirming that the raw power at contact remains intact. What has changed is the angle and direction of contact, which produces ground balls and flat line drives instead of the pulled fly balls that leave the yard. His mechanics explain much of the drought.

But not all of it. Even after adjusting for his degraded 2026 Statcast profile, the Cox model assigns only a 3.96% probability to a 180-PA homer-less stretch. This drought is simultaneously explained and anomalous. The profile shift accounts for a significant portion of the outcome, but the extreme duration still represents a statistically rare event by any reasonable standard.

The survival analysis framework proves well suited to this problem. Its native handling of right-censored observations, its ability to incorporate time-invariant covariates, and its interpretable

hazard ratio output make it a practical tool for player evaluation contexts where “when will this end” is as important as “will this end.”

Assumptions

This analysis assumes proportional hazards, where the covariate effects on drought duration are constant over time, which was confirmed via Schoenfeld residuals (all $p > 0.05$). Episodes are treated as independent observations constructed within each season independently, avoiding cross-season conflation of offseasons adjustments with in-season variance. Statcast batted ball metrics are assumed to be accurately measured, and a hit coordinate threshold of $hc_x < 125$ is used as the standard convention for classifying pulled balls for right-handed batters.

Limitations

This analysis is built on a single player’s 159 drought episodes, limiting direct generalizability to other hitters without refitting on a broader population. The standard Cox model does not account for correlation between consecutive episodes from the same player, but a frailty or gap-time model would address this more rigorously. Covariates are averaged across each episode instead of modeled as time-varying, smoothing over within-drought fluctuations in launch angle or barrel rate. Survival estimates beyond 100 plate appearances carry meaningful uncertainty due to sparse historical data in the tail, and the 2026 covariate profile is a snapshot through May 14 that will shift as the season progresses.

Challenges

The primary technical challenge was deriving the barrel indicator from the `launch_speed_angle` classification field after the expected barrel column was absent from the pybaseball output, requiring careful null handling to avoid misclassifying non-contact plate appearances. Episode construction involved deliberate design decisions around within-season scoping, censoring logic for active droughts, and cumulative sum grouping with downstream modeling implications. Multicollinearity required dropping expected wOBA ($r = 0.76$ with barrel rate) from the covariate set, and the sparse tail of the survival distribution limited the precision of anomaly quantification at extreme drought lengths.

Future Use and Additional Applications

The most impactful extension would be refitting the Cox model on a broad population of power hitters, enabling generalizable hazard ratio estimates and cross-player drought comparisons. Time-varying covariates would capture mechanical shifts that occur mid-drought, and a shared frailty model would explicitly account for correlation between episodes from the same player. A nearest-neighbor feature identifying historical droughts with similar covariate profiles, and how those resolved, would add narrative value for broadcast, coaching, and front office contexts. An interactive application allowing analysts to input any player's Statcast metrics and generate a personalized survival curve would make the methodology accessible to non-technical audiences.

Recommendations

Launch angle should be monitored as the primary leading indicator. Exit velocity is stable, so a sustained return toward Tatis' career average of 13.5 degrees is the strongest early signal that the

drought is approaching resolution. The drought should be treated as partially mechanical rather than purely random variance, as the survival model confirms his 2026 profile meaningfully extends expected drought duration beyond his historical baseline. Plate appearance count rate instead of calendar time is the more meaningful measurement unit for tracking drought progress, and predicted survival curves should be recalculated periodically as 2026 data accumulates, and his profile potentially shifts.

Implementation Plan

Deployment would be in four phases. First, automating the pybaseball data pull on a daily schedule with episode construction logic built into a reusable Python module. Second, serializing the fitted Cox model as a lightweight API endpoint that accepts a player's covariate profile and returns a predicted survival curve. Third, building an app as a front-end with an interactive comparable drought feature for analyst use. Fourth, expanding the dataset to a full population of power-hitters and refitting the model to produce generalizable estimates and enable cross-player comparisons.

Ethical Assessment

This analysis uses publicly available aggregate performance data and makes no claims about player effort, character, or injury. The results are strictly descriptive and probabilistic and should be presented with appropriate uncertainty. Any population-level extension should be examined for representational bias introduced by power-metric filtering criteria. Survival probability outputs should not be used as betting or investment tools, and any commercial deployment

should make model inputs, assumptions, and limitations transparent to end users rather than presenting outputs as black-box predictions.

References

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Appendix

Cox Model Full Output

The table below presents the complete Cox Proportional Hazards model summary including coefficients, hazards ratios, standard errors, 95% confidence intervals, and p-values for all six covariates.

Covariate	Coef	HR	SE	Lower 95%	Upper 95%	z	p	Sig.
barrel_rate	3.854	47.17	0.564	2.747	4.960	6.828	<0.001	Yes
mean_launch_angle	0.031	1.031	0.010	0.010	0.051	2.958	0.003	Yes
mean_exit_velo	0.030	1.031	0.019	-0.007	0.067	1.600	0.110	No
mean_plate_z	0.351	1.420	0.386	-0.407	1.108	0.908	0.364	No
pull_rate	0.246	1.278	0.453	-0.643	1.134	0.542	0.588	No
mean_plate_x	-0.060	0.942	0.391	-0.826	0.706	-0.154	0.878	No

Model fit statistics:

model	lifelines.CoxPHFitter	Concordance	0.773
duration col	'duration'	Partial AIC	1145.238
event col	'event_observed'	log-likelihood ratio test	110.355 on 6 df
penalizer	0.100	-log2(p) of ll-ratio test	68.980
l1 ratio	0.000		
baseline estimation	breslow		
number of observations	158		
number of events observed	151		
partial log-likelihood	-566.619		
time fit was run	2026-05-15 17:26:32 UTC		

Proportional Hazards Assumption (Schoenfeld Residuals):

Covariate	Test Statistic	p-value
barrel_rate	1.335	0.248
mean_exit_velo	2.203	0.138
mean_launch_angle	0.595	0.441
mean_plate_x	0.692	0.406
mean_plate_z	0.994	0.319
pull_rate	1.386	0.239

No covariates violated the proportional hazards assumption (all $p > 0.05$).